Practical No. 6

Aim: Write a JAX-WS web service to perform the following operations. Define a Servlet / JSP that consumes the web service.

Theory:

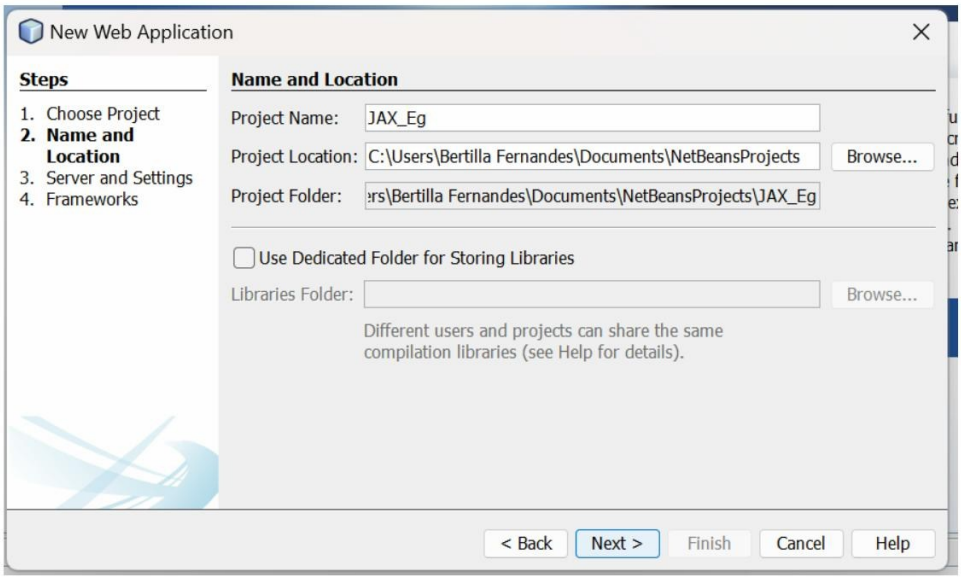
JAX-WS stands for Java API for XML Web Services. It's a Jakarta EE API for building web services and clients that use XML to communicate. JAX-WS allows developers to write both message-oriented and RPC-oriented web services. It uses technologies defined by the World Wide Web Consortium (W3C), including HTTP, SOAP, and Web Service Description Language (WSDL). JAX-WS is platform independent, so a JAX-WS client can access a web service that is not running on the Java platform.

Code:

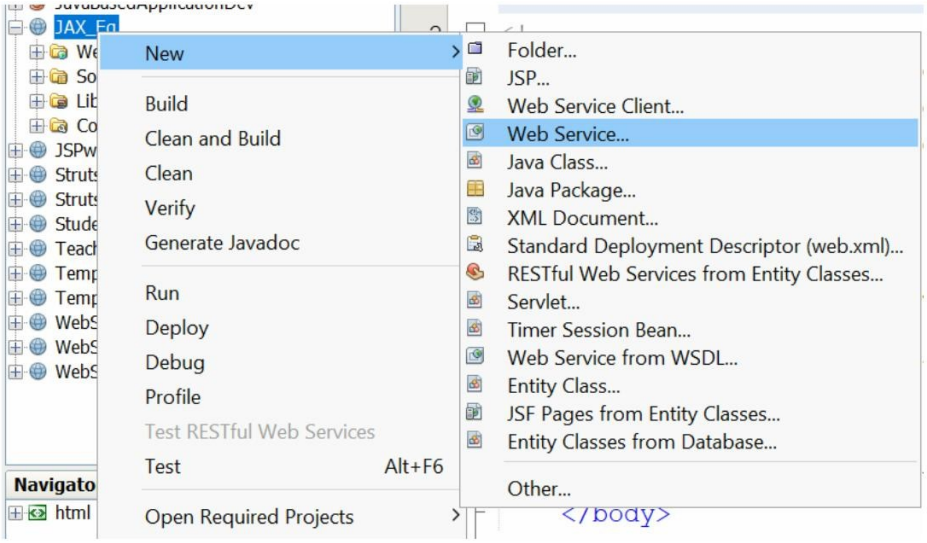
Steps:

1. Go to File -> New Project. Select Java Web in categories and Web Application in Projects. Click on Next to create web based project.

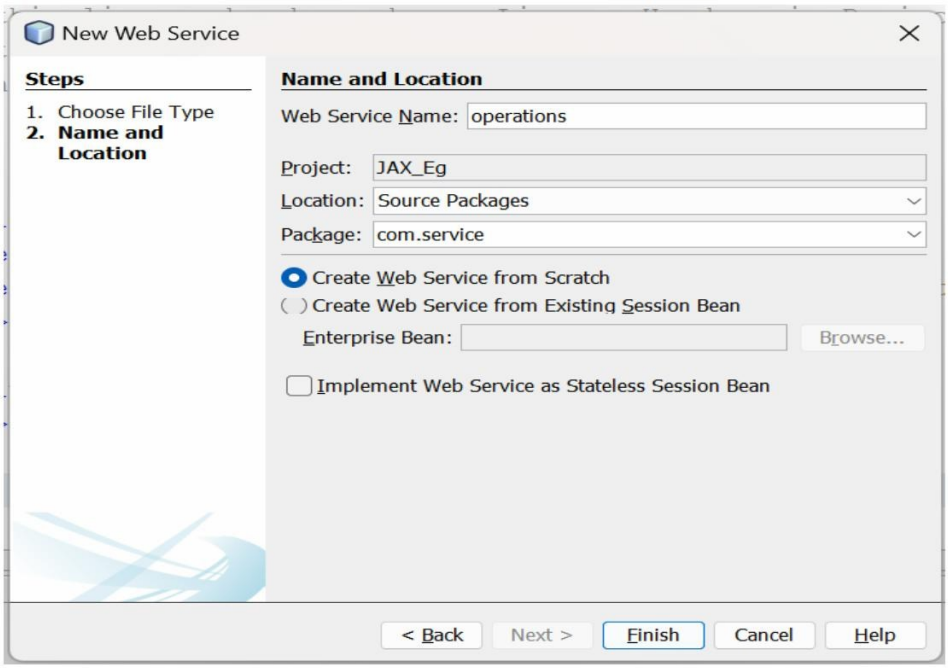
2.Enter a project name “JAX_Eg” whatever you want and then click on Next. On next page click Finish.



3.Create web service. Right click on Project -> New -> Web Service



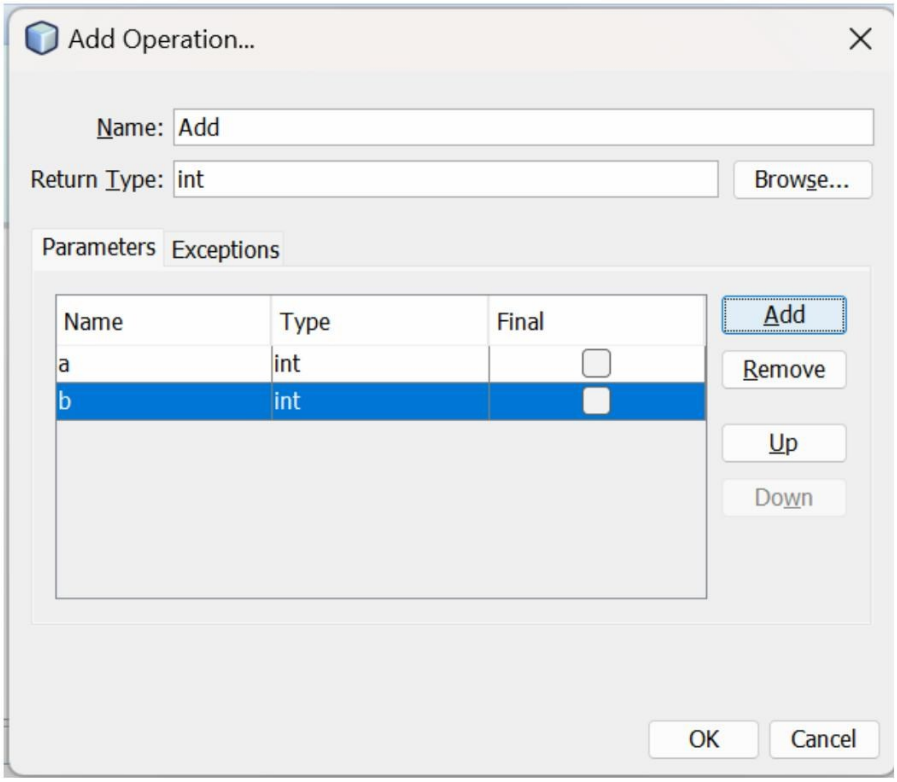
4.Enter a Web Service Name and package name and then click on Finish to create a Web Service.



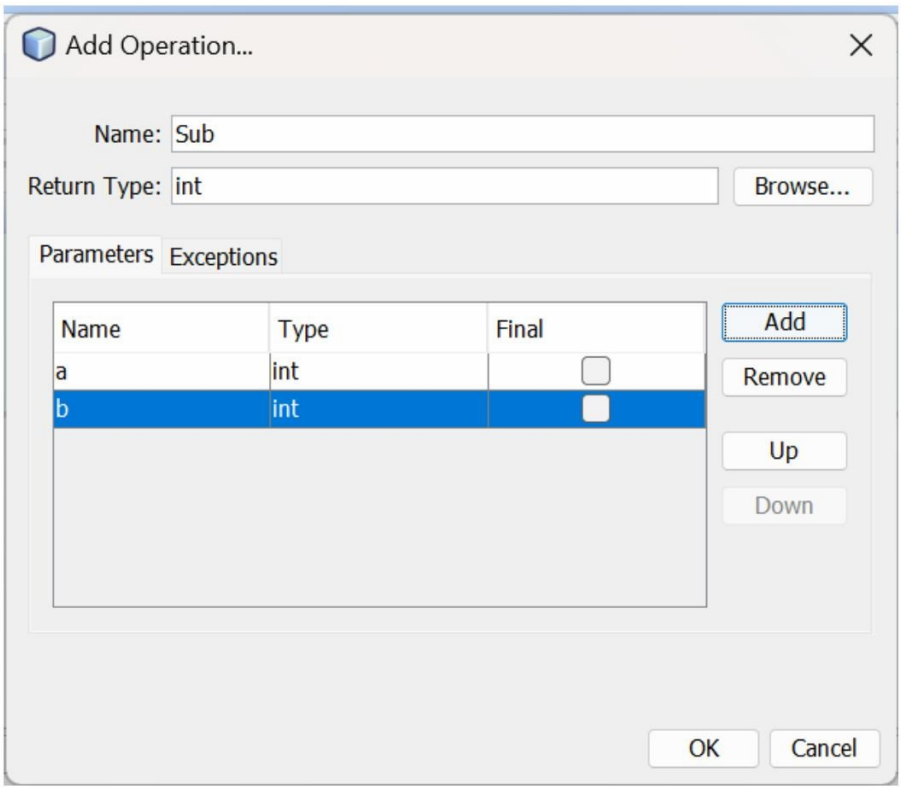
5.Open operations.java file by double click on it, So that we can add four operation that will perform the operations respectively. Go to design mode by click on Design.

Click on Add Operation to add operation.

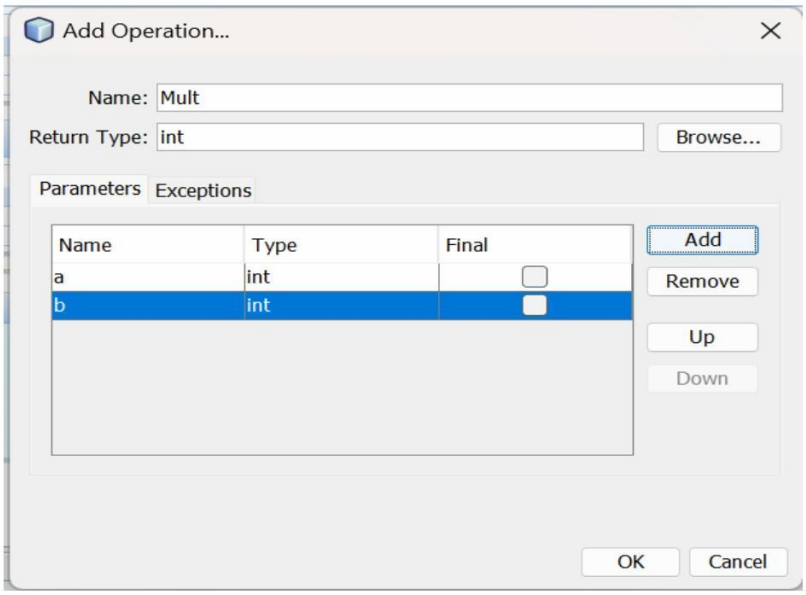
Give Operation name Add and return type as int. So this method will return value in int data type. After that click on Add button to give parameters for method. Give its name as a and type as int. Add another parameter, Give its name as b and type as int. Then click on OK button.



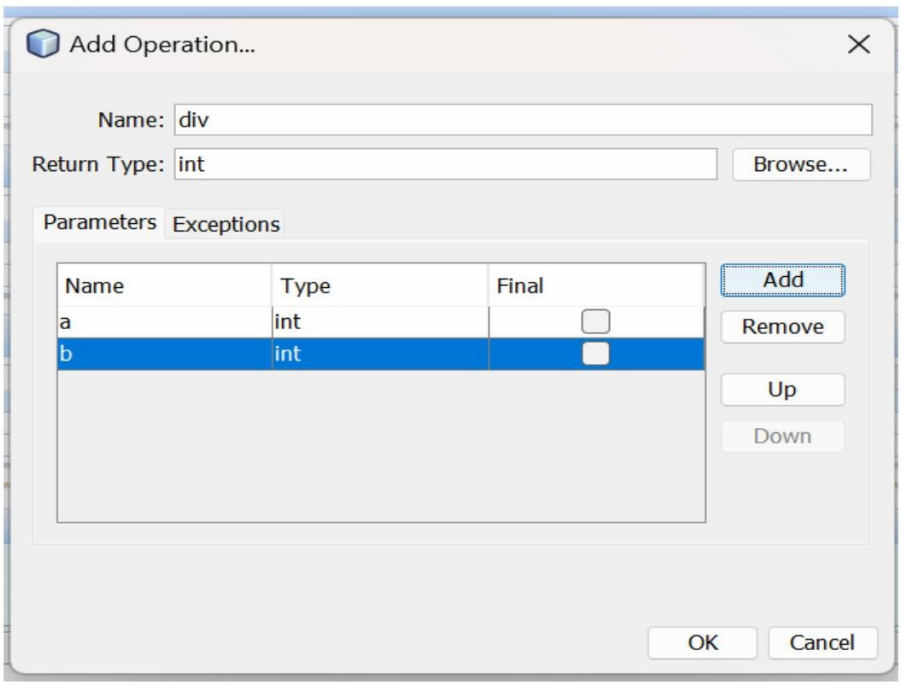
6.Give Operation name Sub and return type as int. So this method will return value in int data type. After that click on Add button to give parameters for method. Give its name as a and type as int. Add another parameter, Give its name as b and type as int. Then click on OK button.



7.Give Operation name Mult and return type as int. So this method will return value in int data type. After that click on Add button to give parameters for method. Give its name as a and type as int. Add another parameter, Give its name as b and type as int. Then click on OK button.



8.Give Operation name div and return type as int. So this method will return value in int data type. After that click on Add button to give parameters for method. Give its name as a and type as int. Add another parameter, Give its name as b and type as int. Then click on OK button.



9.Create the following code perform the arithemetic operations using JAX-WS.

operations.java

```
/*
 * To change this license header, choose License Headers in Project Properties.
 * To change this template file, choose Tools | Templates
 * and open the template in the editor.
 */
package com.service;

import javax.jws.WebService;
import javax.jws.WebMethod;
import javax.jws.WebParam;

/**
 *
 * @author Bertilla Fernandes
 */
@WebService(serviceName = "operations")
public class operations {

    /**
     * Web service operation
     */
    @WebMethod(operationName = "Add")
    public int Add(@WebParam(name = "a") int a, @WebParam(name = "b") int b) {
        //TODO write your implementation code here:
        int c=a+b;
        return c;
    }
```

```
    /**
     * Web service operation
     */
    @WebMethod(operationName = "Sub")
    public int Sub(@WebParam(name = "a") int a, @WebParam(name = "b") int b) {
        //TODO write your implementation code here:
        int c=a-b;
        return c;
    }

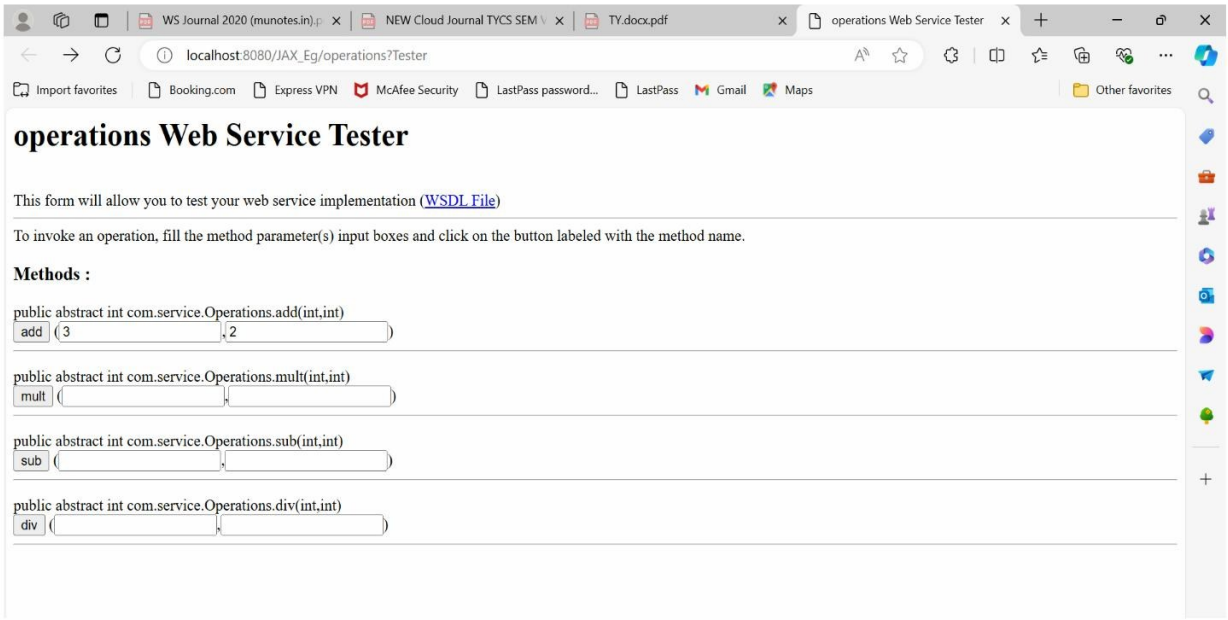
    /**
     * Web service operation
     */
    @WebMethod(operationName = "Mult")
    public int Mult(@WebParam(name = "a") int a, @WebParam(name = "b") int b) {
        //TODO write your implementation code here:
        int c=a*b;
        return c;
    }

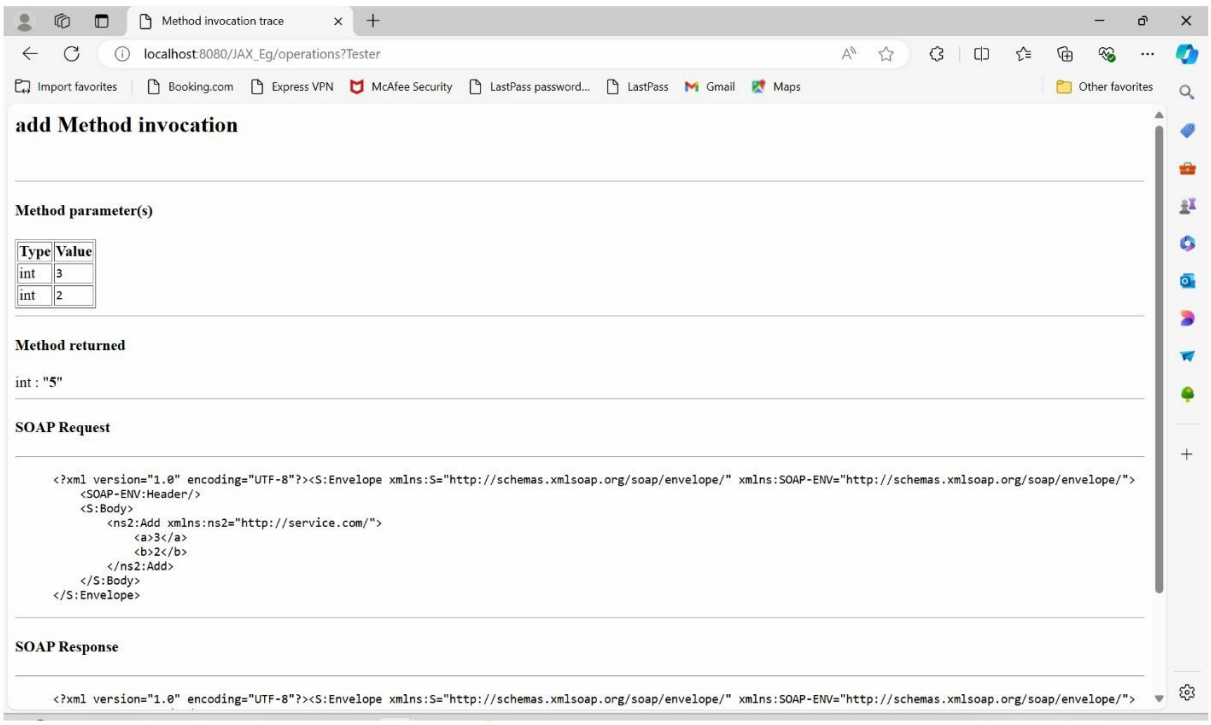
    /**
     * Web service operation
     */
    @WebMethod(operationName = "div")
    public int div(@WebParam(name = "a") int a, @WebParam(name = "b") int b) {
        //TODO write your implementation code here:
        int c=a/b;
        return c;
    }
}
```

10.Now right click on project name and click on Clean and Build your project.

11.Next right click on project name and click on Deploy to deploy your project.

12.Test your web service. Right click on the webservice filename and select the “Test Web Service”.





13.Now to consume this web service, Define a JSP file as “newjsp” and add the following code:

newjsp.jsp

```
<%--
    Document   : newjsp
    Created on : 16 Jul, 2024, 1:45:19 AM
    Author     : Bertilla Fernandes
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
    <head>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
        <title>JSP Page</title>
    </head>
    <body>
        <%
        com.service.operations a1=new com.service.operations();
        int d=a1.Add(5, 10);
        int e=a1.Mult(10, 20);
        int f=a1.Sub(30, 10);
        int g=a1.div(100, 50);

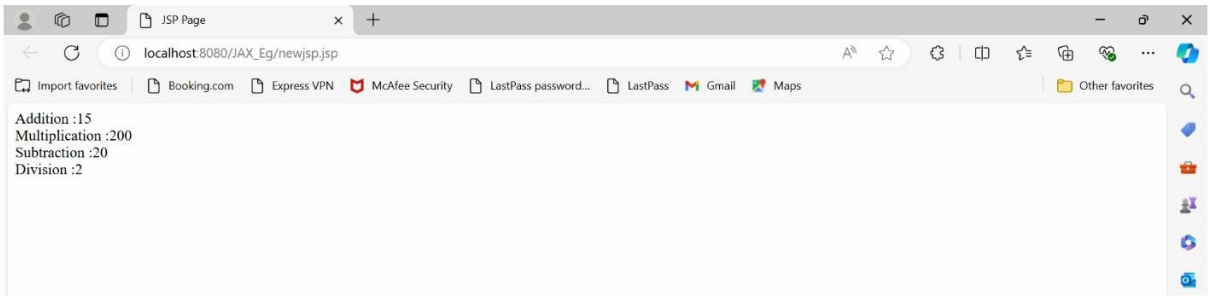
        out.println("Addition :" + d+"<br>");
        out.println("Multiplication :" + e+"<br>");
        out.println("Subtraction :" + f+"<br>");
        out.println("Division :" + g);

        %>
    </body>
</html>
```

14.Now right click on project name and click on Clean and Build your project.

15. Run the JAX_Eg project by running the JSP file “newjsp.jsp”

Output:



References:

1. https://docs.oracle.com/javaee/6/tutorial/doc/bnayn.html
2. https://netbeans.apache.org/tutorial/main/kb/docs/websvc/client/